

CSCI 5244: Homework Set 2

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1. Recall Pauli X,Y,Z matrices: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(a) Show that for any $j \in \{1, 2, 3\}$

$$e^{-i\theta\sigma_j} = I \cos(\theta) - i\sigma_j \sin(\theta)$$

Begin solution

It is known that e^x can expand into the Taylor series

$$e^x = \sum_{n=0}^{\infty} \frac{(-x)^k}{k!} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3} + \dots$$

For Pauli matrices X, Y, Z, when powers are even they all become the identity matrix I :

$$X^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$Y^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$Z^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

If we take the cubic power fo these matricse they will all turn back into themselves:

$$\begin{aligned} X^3 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ Y^3 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ Z^3 &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

So putting in $x = i\theta\sigma_j$ we have

$$\begin{aligned} e^{-i\theta\sigma_j} &= 1 - i\theta\sigma_j + \frac{(i\theta\sigma_j)^2}{2!} - \frac{(i\theta\sigma_j)^3}{3!} + \frac{(i\theta\sigma_j)^4}{4!} \\ &= 1 - i\theta\sigma_j - I \frac{\theta^2}{2!} + i\sigma_j \frac{\theta^3}{3!} + I \frac{\theta^4}{4!} \dots \end{aligned}$$

using how the $X^2, Y^2, Z^2 = I$ and $i^2 = -1, i^3 = -i, i^4 = 1$, etc.

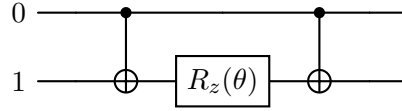
Recall that $i^2 = -1, i^3 = -i, i^4 = 1$. The Taylor series expansion of sin and cos are

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \text{ and } \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

So

$$\begin{aligned} e^{-i\theta\sigma_j} &= \left(1 - I\frac{\theta^2}{2!} + I\frac{\theta^4}{4!} - \dots\right) - i\left(\theta - \sigma_j\frac{\theta^3}{3!} + \sigma_j\frac{\theta^5}{5!} - \dots\right) \\ &= I \cos \theta - i\sigma_j \sin \theta \end{aligned}$$

- (b) Consider a system of two qubits. Denote Z_0 and Z_1 the Pauli Z matrices on qubits 0 and 1, respectively. Show that the unitary transformation $U(\theta) = e^{-i\theta Z_0 Z_1}$ can be implemented by the following circuit with $R_z(\theta) = e^{-i\theta\sigma_3}$



Begin solution

We want to show that this circuit implements the unitary transformation $U(\theta) = e^{-i\theta Z_0 Z_1}$.

Since $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$ for any basis state $|x_0 x_1\rangle$

$$Z_0 Z_1 |x_0 x_1\rangle = (-1)^{x_0 \otimes x_1} |x_0 x_1\rangle$$

Using $R_z(\theta) = I \cos(\theta) - i\sigma_3 \sin(\theta)$ from part (a), we get $R_z(\theta)|0\rangle = e^{-i\theta}|0\rangle$ and $R_z(\theta)|1\rangle = e^{i\theta}|1\rangle$. Below is the trace of the basis state through each gate:

Input	After CNOT	After $R_z(\theta)$	After CNOT
$ 00\rangle$	$ 00\rangle$	$e^{-i\theta} 00\rangle$	$e^{-i\theta} 00\rangle$
$ 01\rangle$	$ 01\rangle$	$e^{i\theta} 01\rangle$	$e^{i\theta} 00\rangle$
$ 10\rangle$	$ 11\rangle$	$e^{i\theta} 11\rangle$	$e^{i\theta} 10\rangle$
$ 11\rangle$	$ 10\rangle$	$e^{-i\theta} 10\rangle$	$e^{-i\theta} 11\rangle$

In every case the circuit output matches $e^{-i\theta Z_0 Z_1}$ exactly, thus this circuit implements the unitary transformation $U(\theta) = e^{-i\theta Z_0 Z_1}$.

2. Amplitude Amplification

Given a unitary operator U on a system of n qubits:

$$U|o^{\otimes n}\rangle = \cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle$$

where $\langle \Psi_0 | \Psi_0 \rangle = \langle \Psi_1 | \Psi_1 \rangle = 1$ and $\langle \Psi_0 | \Psi_1 \rangle = 0$. Define $S_1 = I - 2|\Psi_1\rangle\langle \Psi_1|$, and $S_i = I - 2|0^{\otimes n}\rangle\langle 0^{\otimes n}|$, and $Q = -US_iU^{-1}S_1$.

- (a) Show that

$$\begin{aligned} Q|\Psi_1\rangle &= \cos 2\theta |\Psi_1\rangle - \sin 2\theta |\Psi_0\rangle \\ Q|\Psi_0\rangle &= \sin 2\theta |\Psi_1\rangle + \cos 2\theta |\Psi_0\rangle \end{aligned}$$

Implies the two-dimensional space spanned by $|\Psi_0\rangle$ and $|\Psi_1\rangle$ is invariant under operator Q .

Begin solution

Applying $S_1 |\Psi_1\rangle$

$$\begin{aligned} S_1 |\Psi_1\rangle &= (I - 2 |\Psi_1\rangle \langle \Psi_1|) |\Psi_1\rangle \\ &= I |\Psi_1\rangle - 2 |\Psi_1\rangle \langle \Psi_1 | \Psi_1 \rangle \\ &= -|\Psi_1\rangle \end{aligned}$$

Applying $S_1 |\Psi_0\rangle$

$$\begin{aligned} S_1 |\Psi_0\rangle &= (I - 2 |\Psi_1\rangle \langle \Psi_1|) |\Psi_0\rangle \\ &= I |\Psi_0\rangle - 2 |\Psi_1\rangle \langle \Psi_1 | \Psi_0 \rangle \\ &= |\Psi_1\rangle \end{aligned}$$

Therefore we can conclude that S_1 is a reflection operator when applying S_1 to $|\Psi_1\rangle$ and $|\Psi_0\rangle$.

Now, I will walk through $|\Psi_1\rangle$ first and then come back to $|\Psi_0\rangle$.

Applying U^{-1}

It is defined above that $U |0^{\otimes n}\rangle = \cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle$ which is in the basis $\{|\Psi_0\rangle, |\Psi_1\rangle\}$. We only know what U does to $|0^{\otimes n}\rangle$, but since U is unitary, we can find an orthogonal compliment I will call $|\Phi\rangle$ to map onto an orthonormal basis $\{U |0^{\otimes n}\rangle, |\Phi\rangle\}$.

The orthogonal vector to $U |0^{\otimes n}\rangle$ is the equation $-\sin \theta |\Psi_0\rangle + \cos \theta |\Psi_1\rangle$ because

$$\cos \theta (-\sin \theta) + \sin \theta (\cos \theta) = 0$$

Which is their inner product.

Using the fact that if $\{a, b\}$ is an orthonormal basis, then any vector can be written $|v\rangle = c_a |a\rangle + c_b |b\rangle$, and the coefficients can be extracted by taking an inner product with the corresponding basis vector.

$$\langle a | v \rangle = c_a \langle a | a \rangle + c_b \langle a | b \rangle = c_a (1) + c_b (0) = c_a$$

Applying this to basis $\{U |0^{\otimes n}\rangle, |\Phi\rangle\}$ for $|\Psi_1\rangle$

$$\begin{aligned} |\Psi_1\rangle &= \langle U |0^{\otimes n} | \Psi_1 \rangle \cdot U |0^{\otimes n}\rangle + \langle \Phi | \Psi_1 \rangle \cdot |\Phi\rangle \\ \langle U |0^{\otimes n} | \Psi_1 \rangle &= \cos \theta \langle \Psi_0 | \Psi_1 \rangle + \sin \theta \langle \Psi_1 | \Psi_1 \rangle = \sin \theta \\ \langle \Phi | \Psi_1 \rangle &= -\sin \theta \langle \Psi_0 | \Psi_1 \rangle + \cos \theta \langle \Psi_1 | \Psi_1 \rangle = \cos \theta \end{aligned}$$

Apply the negative given to us by $S_1 |\Psi_1\rangle = -|\Psi_1\rangle$

$$U^{-1} S_1 |\Psi_1\rangle = -(\sin \theta |0^{\otimes n}\rangle + \cos \theta |\phi\rangle)$$

For simplicity's sake, I am leaving out the negative until the very end.

Apply $S_i = I - 2 |0^{\otimes n}\rangle \langle 0^{\otimes n}|$

$$\begin{aligned}
S_i &= \left(I - 2 |0^{\otimes n}\rangle \langle 0^{\otimes n}| \right) \left(\sin \theta |0^{\otimes n}\rangle + \cos \theta |\phi\rangle \right) \\
&= \sin \theta |0^{\otimes n}\rangle + \cos \theta |\phi\rangle - 2 \sin \theta |0^{\otimes n}\rangle \langle 0^{\otimes n}| 0^{\otimes n}\rangle + 2 \cos \theta |0^{\otimes n}\rangle \langle 0^{\otimes n}|\phi\rangle \\
&= \sin \theta 0^{\otimes n} + \cos \theta |\phi\rangle - 2 \sin \theta |0^{\otimes n}\rangle (1) + 2 \cos \theta |0^{\otimes n}\rangle (0) \\
&= -\sin \theta |0^{\otimes n}\rangle + \cos \theta |\phi\rangle
\end{aligned}$$

Apply U

$$\begin{aligned}
US_iU^{-1}S_1|\Psi_1\rangle &= -\sin \theta U|0^{\otimes n}\rangle + \cos \theta |\phi\rangle \\
&= -\sin \theta (\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle) + \cos \theta (-\sin \theta |\Psi_0\rangle + \cos \theta |\Psi_1\rangle) \\
&= -\sin \theta \cos \theta |\Psi_0\rangle + -\sin^2 \theta |\Psi_1\rangle + -\sin \theta \cos \theta |\Psi_0\rangle + \cos^2 \theta |\Psi_1\rangle \\
&= \cos^2 \theta |\Psi_1\rangle - \sin^2 \theta |\Psi_1\rangle - 2 \sin \theta \cos \theta |\Psi_0\rangle \\
&= \cos(2\theta) |\Psi_1\rangle - \sin(2\theta) |\Psi_0\rangle
\end{aligned}$$

Apply both negatives from the equation $S_1 = -|\Psi_1\rangle$ and from the beginning of the equation $Q = -US_iU^{-1}S_1$, which cancels out giving us

$$\cos(2\theta) |\Psi_1\rangle - \sin(2\theta) |\Psi_0\rangle$$

Matching the original equation.

Back to solving $|\Psi_0\rangle$. Recall that

$$\begin{aligned}
S_1|\Psi_0\rangle &= (I - 2 |\Psi_1\rangle \langle \Psi_1|) |\Psi_0\rangle \\
&= I |\Psi_0\rangle - 2 |\Psi_1\rangle \langle \Psi_1|\Psi_0\rangle \\
&= |\Psi_1\rangle
\end{aligned}$$

Applying U^{-1} which was established above to be $-\sin \theta |\Psi_0\rangle + \cos \theta |\Psi_1\rangle$. Map $|\Psi_0\rangle$ to orthonormal basis $\{U|0^{\otimes n}\rangle, |\Phi\rangle\}$

$$\begin{aligned}
|\Psi_0\rangle &= \langle U0^{\otimes n}|\Psi_0\rangle \cdot U|0^{\otimes n}\rangle + \langle \Phi|\Psi_0\rangle \cdot |\Phi\rangle \\
\langle U0^{\otimes n}|\Psi_0\rangle &= \cos \theta \langle \Psi_0|\Psi_0\rangle + \sin \theta \langle \Psi_1|\Psi_0\rangle = \cos \theta \\
\langle \Phi|\Psi_0\rangle &= -\sin \theta (1) + \cos \theta (0) = -\sin \theta \\
|\Psi_0\rangle &= \cos \theta U|0^{\otimes n}\rangle - \sin \theta |\Phi\rangle
\end{aligned}$$

Applying S_i

$$\begin{aligned}
S_i &= \left(I - 2 |0^{\otimes n}\rangle \langle 0^{\otimes n}| \right) \left(\cos \theta |0^{\otimes n}\rangle - \sin \theta |\phi\rangle \right) \\
&= \cos \theta |0^{\otimes n}\rangle - \sin \theta |\phi\rangle - 2 \cos \theta |0^{\otimes n}\rangle \langle 0^{\otimes n}| 0^{\otimes n}\rangle + 2 \sin \theta |0^{\otimes n}\rangle \langle 0^{\otimes n}|\phi\rangle \\
&= \cos \theta |0^{\otimes n}\rangle - \sin \theta |\phi\rangle - 2 \cos \theta |0^{\otimes n}\rangle (1) + 2 \sin \theta |0^{\otimes n}\rangle (0) \\
&= -\cos \theta |0^{\otimes n}\rangle - \sin \theta |\phi\rangle
\end{aligned}$$

Apply U

$$\begin{aligned}
US_i U^{-1} S_1 |\Psi_0\rangle &= -\cos \theta |0^{\otimes n}\rangle - \sin \theta |\phi\rangle \\
&= -\cos \theta (\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle) - \sin \theta (-\sin \theta |\Psi_0\rangle + \cos \theta |\Psi_1\rangle) \\
&= -\cos^2 \theta |\Psi_0\rangle - \sin \theta \cos \theta |\Psi_1\rangle + \sin^2 \theta |Psi_0\rangle - \sin \theta \cos \theta |\Psi_1\rangle \\
&= \sin^2 \theta |\Psi_0\rangle - \cos^2 \theta |\Psi_0\rangle - 2 \sin \theta \cos \theta |\Psi_1\rangle
\end{aligned}$$

Apply “ $-$ ” operator

$$\begin{aligned}
&= \cos^2 \theta |\Psi_0\rangle - \sin^2 \theta |\Psi_0\rangle + 2 \sin \theta \cos \theta |\Psi_1\rangle \\
&= \cos(2\theta) |\Psi_0\rangle + \sin(2\theta) |\Psi_1\rangle
\end{aligned}$$

Which is equal to Q defined above.

(b) Show that

$$Q^k U |0^{\otimes n}\rangle = \cos(2k+1)\theta |\Psi_0\rangle + \sin(2k+1)\theta |\Psi_1\rangle, k \in \mathbb{Z}_+$$

Suppose θ is very small. If $|\Psi_1\rangle$ is the desired result, the above relation implies that the probability of obtaining the desired result increases quadratically with the number of times Q is applied.

Begin solution

This is a proof by induction. Idk if I'm supposed to say that, but that's the technique I am using.

We can establish a base case of $k = 0$

$$\begin{aligned}
Q^0 U |0^{\otimes n}\rangle &= \cos(2(0)+1)\theta |\Psi_0\rangle + \sin(2(0)+1)\theta |\Psi_1\rangle \\
U |0^{\otimes n}\rangle &= \cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle
\end{aligned}$$

Assume that this holds for some $k+1$

$$\begin{aligned}
Q^{k+1} U |0^{\otimes n}\rangle &= \cos(2(k+1)+1)\theta |\Psi_0\rangle + \sin(2(k+1)+1)\theta |\Psi_1\rangle \\
&= \cos(2k+3)\theta |\Psi_0\rangle + \sin(2k+3)\theta |\Psi_1\rangle
\end{aligned}$$

Q^{k+1} can also be represented by $Q^1 Q^k$ and since we already know Q^k we can expand this as

$$Q^{k+1} = Q(Q^k U |0^{\otimes n}\rangle) = Q(\cos(2k+1)\theta |\Psi_0\rangle + \sin(2k+1)\theta |\Psi_1\rangle)$$

Distribute Q by linearity which was proven in 2(a) using

$$\begin{aligned}
Q |\Psi_1\rangle &= \cos 2\theta |\Psi_1\rangle - \sin 2\theta |\Psi_0\rangle \\
Q |\Psi_0\rangle &= \sin 2\theta |\Psi_1\rangle + \cos 2\theta |\Psi_0\rangle
\end{aligned}$$

Applying to the aforementioned formula

$$= \cos(2k+1)\theta \left(\sin(2\theta) |\Psi_1\rangle + \cos(2\theta) |\Psi_0\rangle \right) \\ + \sin(2k+1)\theta \left(\cos(2\theta) |\Psi_1\rangle - \sin(2\theta) |\Psi_0\rangle \right)$$

Expanded this is (really long)

$$= \cos(2k+1)\theta \sin(2\theta) |\Psi_1\rangle + \cos(2k+1)\theta \cos(2\theta) |\Psi_0\rangle \\ + \sin(2k+1)\theta \cos(2\theta) |\Psi_1\rangle - \sin(2k+1)\theta \sin(2\theta) |\Psi_0\rangle$$

Grouping all $|\Psi_0\rangle$ together results in

$$\left[\cos(2k+1)\theta \cos(2\theta) - \sin(2k+1)\theta \sin(2\theta) \right] |\Psi_0\rangle$$

Applying trig identity $\cos(A)\cos(B) - \sin(A)\sin(B) = \cos(A+B)$

$$\cos(2k+1+2)\theta = \cos(2k+3)\theta |\Psi_0\rangle$$

Grouping all $|\Psi_1\rangle$ together results in

$$\left[\cos(2k+1)\theta \sin(2\theta) + \sin(2k+1)\theta \cos(2\theta) \right] |\Psi_1\rangle$$

Applying trig identity $\cos(A)\sin(B) + \sin(A)\cos(B) = \sin(A+B)$

$$\sin(2+2k+1)\theta = \sin(2k+3)\theta |\Psi_1\rangle$$

This proves that the claim holds true for $k+1$.

Thus the equation

$$Q^k U |O^{\otimes n}\rangle = \cos(2k+1)\theta |\Psi_0\rangle + \sin(2k+1)\theta |\Psi_1\rangle, k \in \mathbb{Z}_+$$

holds true by induction.

3. Consider the quantum Fourier transformation

$$QFT_{2^2} |x\rangle = \frac{1}{\sqrt{2^2}} \sum_{y=0}^{2^2-1} e^{i \frac{2\pi xy}{2^2}} |y\rangle = \frac{1}{\sqrt{2^2}} (|0\rangle + e^{i2\pi(.x_0)} |1\rangle) (|0\rangle + e^{i2\pi(.x_1x_0)} |1\rangle)$$

(a) Recall the notation in lecture 3: $x \equiv x_1x_0$ and $y \equiv y_1y_0$. Show that

$$\frac{xy}{2^2} \pmod{1} = y_1(.x_0) + y_0(.x_1x_0)$$

Begin solution

We can write out the expansion of $x \equiv x_1x_0 = 2x_1 + x_0$ and similarly $y \equiv y_1y_0 = 2y_1 + y_0$.

Substituting this in for $xy/2^2$

$$\begin{aligned} \frac{(2x_1 + x_0)(2y_1 + y_0)}{4} &= \frac{4x_1y_1 + 2x_1y_0 + 2x_0y_1 + x_0y_0}{4} \\ &= x_1y_1 + \frac{x_1y_0}{2} + \frac{x_0y_1}{2} + \frac{x_0y_0}{4} \end{aligned}$$

Since $x_1, y_1 \in \{0, 1\}$ the term x_1y_1 is an integer and vanishes when applying $\text{mod } 1$

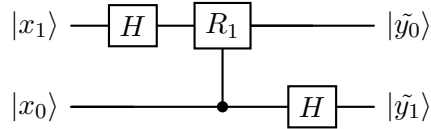
$$\frac{xy}{2} (\text{mod } 1) = \frac{x_1y_0}{2} + \frac{x_0y_1}{2} + \frac{x_0y_0}{4}$$

Expanding the binary fractions $(.x_0)$ and $(.x_1x_0)$

$$y_1(.x_0) + y_0(.x_1x_0) = y_1 \cdot \frac{x_0}{2} + y_0 \left(\frac{x_1}{2} + \frac{x_0}{4} \right) = \frac{x_0y_1}{2} + \frac{x_1y_0}{2} + \frac{x_0y_0}{4}$$

The two expressions agree term by term.

(b) Show that QFT_{2^2} can be computed by the following circuit



where $|\tilde{y}_0\rangle = |0\rangle + e^{i2\pi(.x_1x_0)} |1\rangle$, $|\tilde{y}_1\rangle = |0\rangle + e^{i2\pi(.x_0)}$, and $R_1 = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{2}} \end{pmatrix}$

Begin solution

Apply Hadamard to $|x_0\rangle$

$$H |x_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi x_1} |1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi(.x_1)} |1\rangle)$$

since $e^{i\pi \cdot 0} = 1$ and $e^{i\pi} = -1$.

Gate 2 controlled R_1 on the top wire $|x_1\rangle$. The bottom wire still holds the basis state $|x_0\rangle$, so the joint state is a product and the control bit is definite. R_1 multiplies the $|1\rangle$ amplitude by $e^{i\pi/2}$ when $x_0 = 1$ and acts trivially when $x_0 = 0$; both cases are captured by the phase $e^{i\pi x_0/2}$. The top wire becomes

$$\frac{1}{\sqrt{2}} \left(|0\rangle + e^{i\pi x_1} e^{i\pi x_0/2} |1\rangle \right)$$

Factoring the 2π out of the top gives $\frac{x_1}{2} + \frac{x_0}{4} = (.x_1x_0)$, so the top wire holds

$$|\tilde{y}_0\rangle = |0\rangle + e^{i2\pi(.x_1x_0)} |1\rangle$$

Applying the Hadamard gate on $|x_0\rangle$. The control qubit is unchanged by the bottom wire, so the bottom wire holds $|x_0\rangle$. By the same computation as Gate 1,

$$|\tilde{y}_1\rangle = |0\rangle + e^{i2\pi(.x_0)} |1\rangle$$

Compared with the QFT definition which is

$$QFT_{2^2} |x\rangle = \frac{1}{\sqrt{2^2}} \sum_{y=0}^{2^2-1} e^{i\frac{2\pi xy}{2^2}} |y\rangle = \frac{1}{\sqrt{2^2}} (|0\rangle + e^{i2\pi(.x_0)} |1\rangle)(|0\rangle + e^{i2\pi(.x_1x_0)} |1\rangle)$$

The circuit produces both factors, but with $(.x_1x_0)$ factor on the top wire and $(.x_0)$ on the bottom wire which reverse for what the QFT defines.

However, this is not a problem, because this is a bit-reversal artifact of the QFT circuit which is resolved by either appending a SWAP gate or by reading the output register in reverse, which is what labeling $|\tilde{y}_0\rangle$ (top) and $|\tilde{y}_1\rangle$ (bottom) indicates.

Otherwise, both equations hold true term by term.